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COMPUTATIONAL STATISTICS

EXAMINATION

```
##Install and Load the MASS package
install.packages("MASS")
library(MASS)
```

QUESTION 1

- a. Describe the data set using the one in the package MASS

```
Morphological Measurements on Leptograpsus Crabs
Description
    The crabs data frame has 200 rows and 8 columns,
    describing 5 morphological measurements on 50 crabs,
    each of two colour forms and both sexes, of the species
    Leptograpsus variegatus collected at Fremantle, W.
    Australia.
```

- b. Load the data set and display its structure

CODES

```
data("crabs")
str(crabs)
```

RESULTS

```
data("crabs")
> str(crabs)
'data.frame':   200 obs. of  8 variables:
 $ sp: Factor w/ 2 levels "B", "O": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex: Factor w/ 2 levels "F", "M": 2 2 2 2 2 2 2 2 2 2 ...
 $ index: int  1 2 3 4 5 6 7 8 9 10 ...
 $ FL : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
 $ RW : num  6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
 $ CW : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
 $ BD : num  7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

- c. Frequency tables

CODES

```
table(crabs$sp, crabs$sex)
```

RESULTS

```
> table(crabs$sp,crabs$sex)
      F  M
B  50  50
O  50  50
```

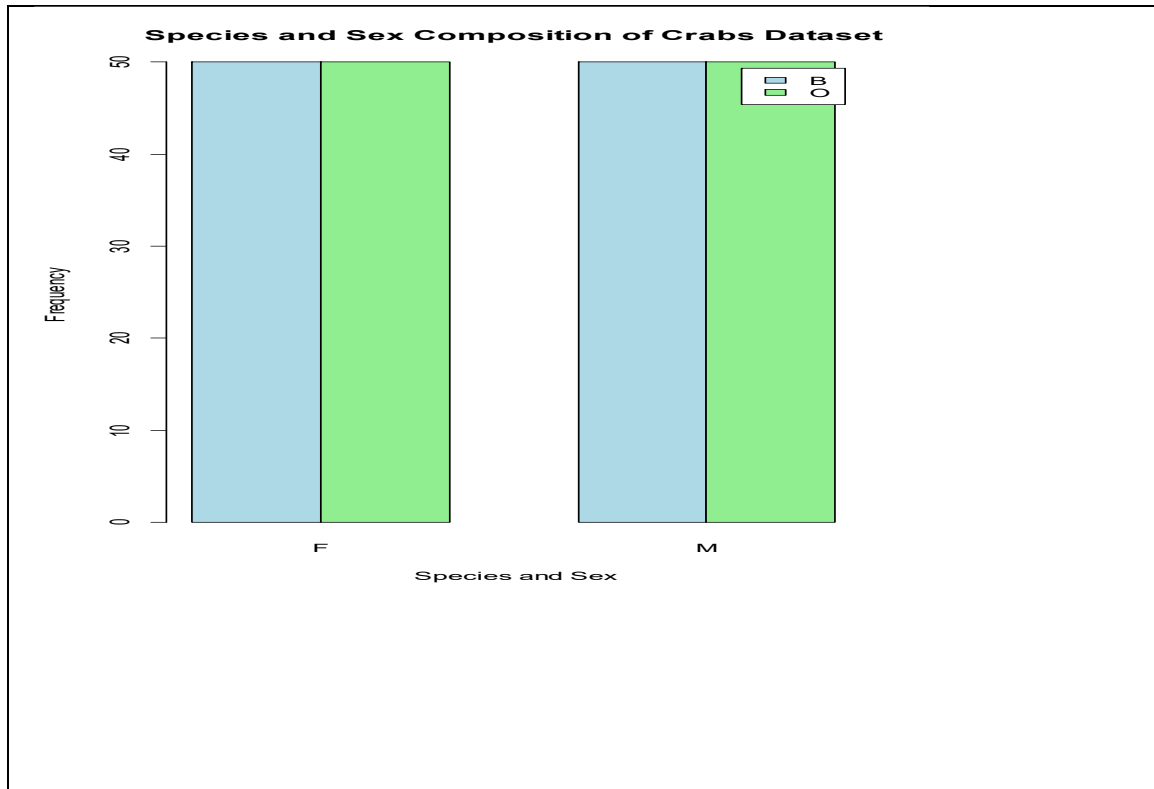
- d. Bar plot for the frequency

CODES

```
Spex<-table(crabs$sp,crabs$sex)
barplot(Spex,
beside = TRUE,
legend = TRUE,
col = c("lightblue", "lightgreen"),
main = "Species and Sex Composition of Crabs Dataset",
xlab = "Species and Sex",
ylab = "Frequency")
```

RESULTS

```
barplot(Spex,
+       beside = TRUE,
+       legend = TRUE,
+       col = c("lightblue", "lightgreen"),
+       main = "Species and Sex Composition of Crabs Dataset",
+       xlab = "Species and Sex",
+       ylab = "Frequency")
```



- e. All four groups are balanced since they equally have 50 observations. There is no bias in accordance with the data.

QUESTION 2

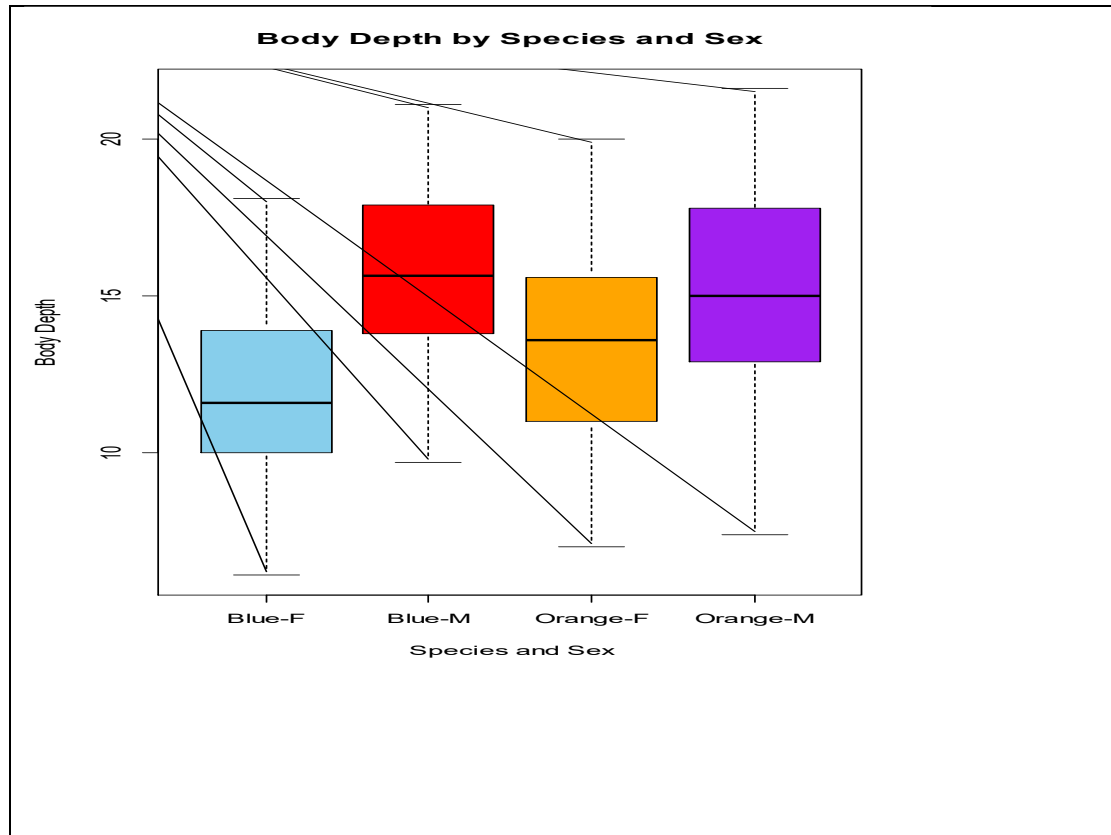
- a. Boxplots of body depth for the four groups

CODES

```
boxplot(BD ~ interaction(sp, sex),
        data = crabs,
        col = c("skyblue", "red", "orange", "purple"),
        main = "Body Depth by Species and Sex",
        xlab = "Species and Sex",
        ylab = "Body Depth",
        names = c("Blue-F", "Blue-M", "Orange-F", "Orange-M"))
```

RESULTS

```
> boxplot(BD ~ interaction(sp, sex),
+         data = crabs,
+         col = c("skyblue", "red", "orange",
+ "purple"),
+         main = "Body Depth by Species and Sex",
+         xlab = "Species and Sex",
+         ylab = "Body Depth",
+         names = c("Blue-F", "Blue-M", "Orange-
F", "Orange-M"))
```



- b. Compare the central tendency and variability between groups
CODES

```
crabs$Group <- interaction(crabs$sp, crabs$sex)
aggregate(BD ~ Group, data = crabs,
  function(x) c(
    mean = mean(x),
    median = median(x),
    sd = sd(x),
    var = var(x),
    IQR = IQR(x),
    min = min(x),
    max = max(x)))
aggregate(BD ~ Group, data = crabs,
  function(x) diff(range(x)))
```

RESULTS

```
crabs$Group <- interaction(crabs$sp, crabs$sex)
aggregate(BD ~ Group, data = crabs,
+   function(x) c(
+     mean = mean(x),
+     median = median(x),
+     sd = sd(x),
+     var = var(x),
```

```

+           IQR = IQR(x),
+           min = min(x),
+           max = max(x)
+       ))
      Group   BD.mean BD.median   BD.sd   BD.var   BD.IQR
BD.min   BD.max
1   B.F 11.816000 11.600000  2.752465  7.576065  3.850000
6.100000 18.100000
2   O.F 15.632000 15.650000  2.752620  7.576914  4.050000
9.700000 21.100000
3   B.M 13.350000 13.600000  3.199888 10.239286  4.600000
7.000000 20.000000
4   O.M 15.324000 15.000000  3.527187 12.441045  4.825000
7.400000 21.600000
> aggregate(BD ~ Group, data = crabs,
+           function(x) diff(range(x)))
      Group   BD
1   B.F 12.0
2   O.F 11.4
3   B.M 13.0
4   O.M 14.2

```

QUESTION 3

a. Histograms of carapace length for each species

CODES

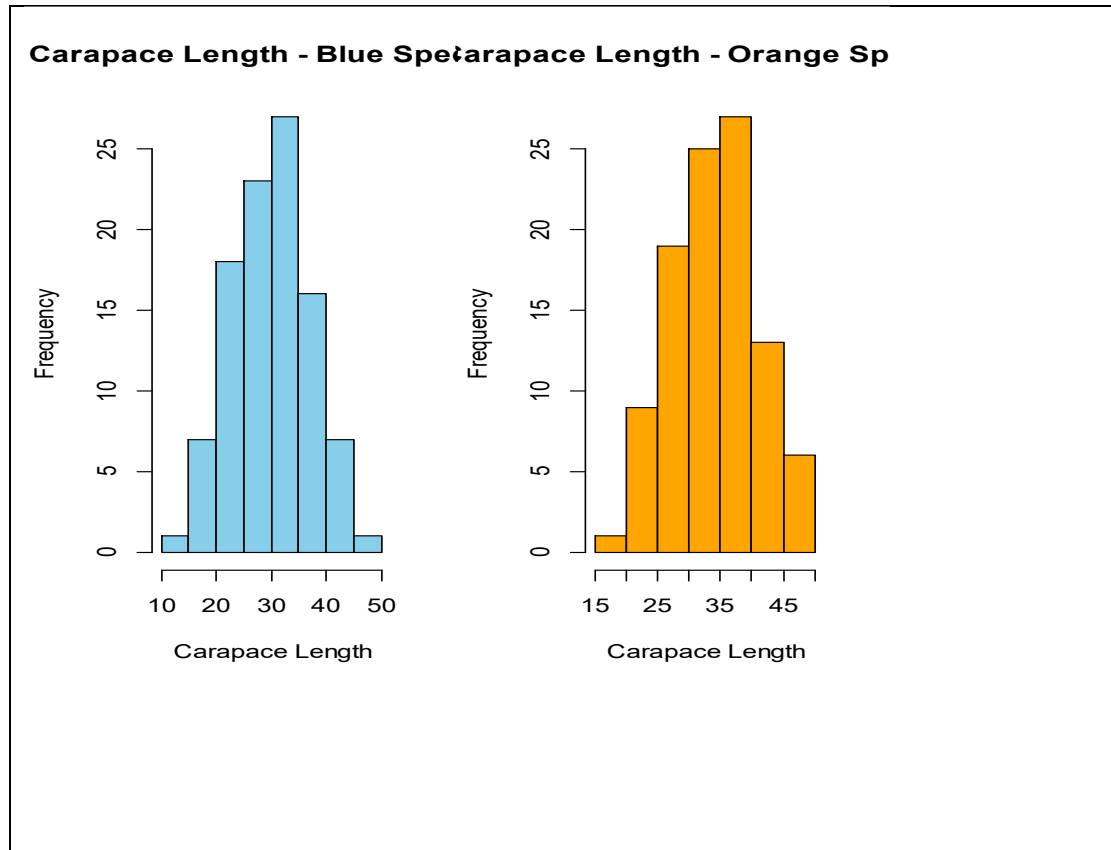
```

par(mfrow = c(1, 2)) # 1 row, 2 columns
hist(crabs$CL[crabs$sp == "B"],
      main = "Carapace Length - Blue Species",
      xlab = "Carapace Length",
      col = "skyblue",
      border = "black")

hist(crabs$CL[crabs$sp == "O"],
      main = "Carapace Length - Orange Species",
      xlab = "Carapace Length",
      col = "orange",
      border = "black")

```

RESULTS



b. Shape of the Distributions

Blue Species (B)

The histogram is approximately symmetric. It is slightly positive (right) skewed, and most values cluster around the center; thus, the mean has gradually decreasing tails.

Orange Species (O)

The histogram is approximately symmetric and is smooth with no extremely long tails. It is also slightly skewed to the right.

To conclude, both species show very similar shapes; neither is strongly skewed, and their tails are moderate.

- For modality, both species show one clear peak. Therefore, both species exhibit unimodal distributions.

QUESTION 4

- a. Mean of each morphological measurement by species and sex

CODES

```
aggregate(FL ~ sp + sex, data = crabs, mean)
aggregate(RW ~ sp + sex, data = crabs, mean)
aggregate(CL ~ sp + sex, data = crabs, mean)
aggregate(CW ~ sp + sex, data = crabs, mean)
aggregate(BD ~ sp + sex, data = crabs, mean)
```

RESULTS

```
aggregate(FL ~ sp + sex, data = crabs, mean)
  sp sex    FL
1  B  F 13.270
2  O  F 17.594
3  B  M 14.842
4  O  M 16.626
> aggregate(FL ~ sp + sex, data = crabs, mean)
  sp sex    FL
1  B  F 13.270
2  O  F 17.594
3  B  M 14.842
4  O  M 16.626
> aggregate(RW ~ sp + sex, data = crabs, mean)
  sp sex    RW
1  B  F 12.138
2  O  F 14.836
3  B  M 11.718
4  O  M 12.262
> aggregate(CL ~ sp + sex, data = crabs, mean)
  sp sex    CL
1  B  F 28.102
2  O  F 34.618
3  B  M 32.014
4  O  M 33.688
> aggregate(CW ~ sp + sex, data = crabs, mean)
  sp sex    CW
1  B  F 32.624
2  O  F 39.036
3  B  M 36.810
4  O  M 37.188
> aggregate(BD ~ sp + sex, data = crabs, mean)
  sp sex    BD
1  B  F 11.816
2  O  F 15.632
3  B  M 13.350
4  O  M 15.324
```

- b. Grouped bar chart to display their means

CODES

```
combine_mean<-aggregate(cbind(FL, RW, CL, CW, BD) ~
  sp + sex,
```

```

        data = crabs,
        mean)
mean_matrix <- as.matrix(combine_mean[, 3:7])
mean_matrix_row <- paste(combine_mean$sp,
combine_mean$sex, sep = "-")
barplot(mean_matrix,
        beside = TRUE,
        col = c("skyblue", "lightblue",
"orange", "gold"),
        legend.text = mean_matrix_row,
        args.legend = list(x = "topright"),
        main = "Mean Morphological Measurements
by Species and Sex",
        xlab = "Measurements",
        ylab = "Mean Value")

```

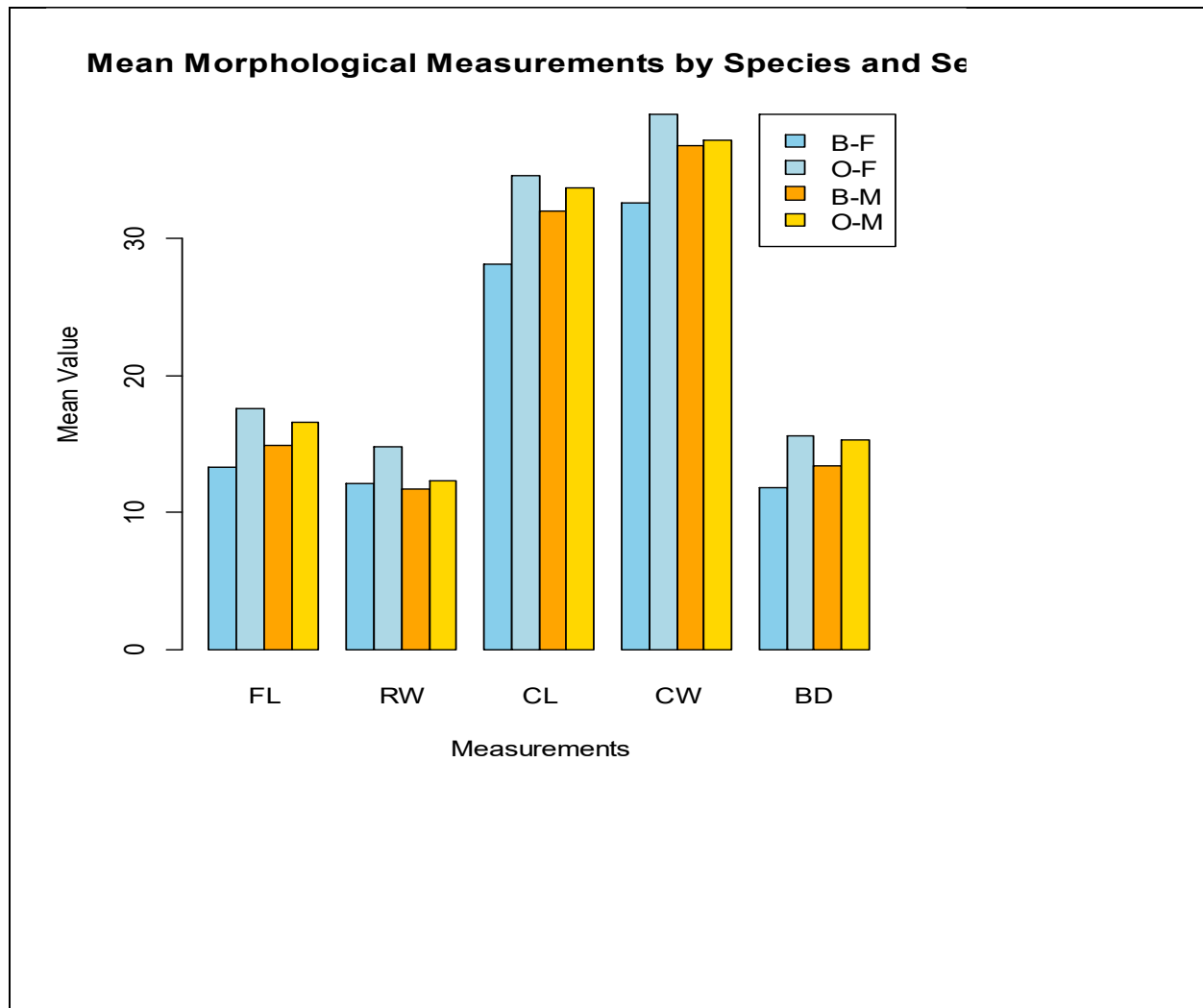
RESULTS

```

combine_mean<-aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,
+   data = crabs,
+   mean)
> combine_mean
  sp sex   FL   RW   CL   CW   BD
1  B  F 13.270 12.138 28.102 32.624 11.816
2  O  F 17.594 14.836 34.618 39.036 15.632
3  B  M 14.842 11.718 32.014 36.810 13.350
4  O  M 16.626 12.262 33.688 37.188 15.324

mean_matrix <- as.matrix(combine_mean[, 3:7])
mean_matrix
      FL   RW   CL   CW   BD
[1,] 13.270 12.138 28.102 32.624 11.816
[2,] 17.594 14.836 34.618 39.036 15.632
[3,] 14.842 11.718 32.014 36.810 13.350
[4,] 16.626 12.262 33.688 37.188 15.324
barplot(mean_matrix,
+   beside = TRUE,
+   col = c("skyblue", "lightblue", "orange", "gold"),
+   legend.text = mean_matrix_row,
+   args.legend = list(x = "topright"),
+   main = "Mean Morphological Measurements by Species and Sex",
+   xlab = "Measurements",
+   ylab = "Mean Value")

```

Greatest Species Difference

Carapace Width (CW) and Carapace Length (CL)

These measurements usually show the clearest separation between Blue and Orange species. Among them, CW often shows the largest species difference.

Least Sex Variation

The frontal lobe (FL) typically shows the smallest difference between males and females. Sex affects body size traits more strongly (like CW and CL), but FL changes less.

QUESTION 5

- Scatter plot matrix pair
CODES

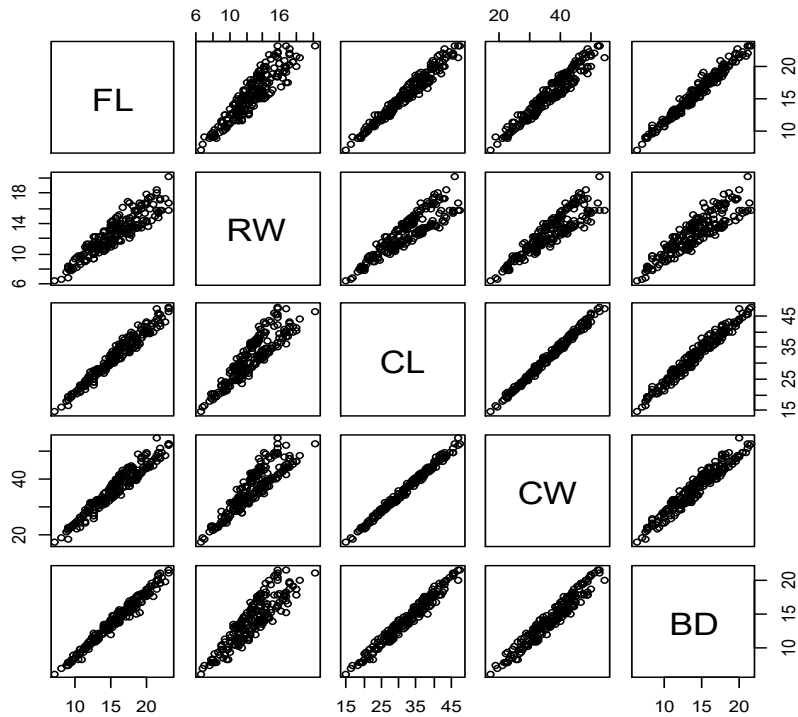
```
pairs(crabs[, c("FL", "RW", "CL", "CW", "BD")],
```

```
main = "Scatter Plot Matrix of Morphological Measurements")
```

RESULTS

```
> pairs(crabs[, c("FL", "RW", "CL", "CW", "BD")],
+       main = "Scatter Plot Matrix of Morphological Measurements")
```

Scatter Plot Matrix of Morphological Measurements



b. Correlation among the variables

CODES

```
cor(crabs[, c("FL", "RW", "CL", "CW", "BD")])
```

```
cor(crabs[, c("FL", "RW", "CL", "CW", "BD")])
      FL      RW      CL      CW      BD
FL 1.0000000 0.9069876 0.9788418 0.9649558 0.9876272
RW 0.9069876 1.0000000 0.8927430 0.9004021 0.8892054
CL 0.9788418 0.8927430 1.0000000 0.9950225 0.9832038
CW 0.9649558 0.9004021 0.9950225 1.0000000 0.9678117
BD 0.9876272 0.8892054 0.9832038 0.9678117 1.0000000
```

From the scatter plot,

Very strong positive relationships are CL and CW, CL and BD, CW and BD. The points are tightly clustered along a straight upward line

Moderate to Strong Positive Relationships are FL with CL, CW, BD, RW. These relationships are still positive but show slightly more scatter.

Summary Statistical Report on the Crabs Dataset

The crabs dataset from the **MASS** contains morphological measurements of 200 crabs classified by species (Blue and Orange) and sex (Male and Female). Five quantitative variables were analysed: frontal lobe size (FL), rear width (RW), carapace length (CL), carapace width (CW), and body depth (BD). The objective was to examine differences in central tendency and variability across groups and to assess relationships among the measurements.

Descriptive statistics indicated clear differences between species and sex groups. Males generally exhibited higher mean values than females across most measurements, reflecting sexual dimorphism. Species differences were most pronounced in carapace length (CL) and carapace width (CW), which showed the largest separation in mean values. In contrast, frontal lobe size (FL) displayed comparatively smaller differences between sexes. Histograms of carapace length by species suggested approximately symmetric and unimodal distributions, with no strong evidence of extreme skewness.

The scatter plot matrix revealed strong positive linear relationships among all five morphological variables. In particular, CL, CW, and BD were very strongly associated, indicating that larger crabs tend to be proportionally larger in all body dimensions. Overall, the analysis demonstrates structured biological variation driven by both species and sex, with overall body size measurements providing the clearest differentiation between groups.